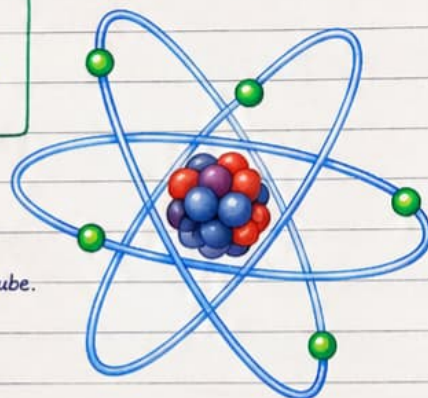




Structure of Atom



An atom is the smallest particle of an element that can exist. It consists of a dense, positively charged nucleus surrounded by negatively charged electrons.



1. DISCOVERY OF SUBATOMIC PARTICLES

- **Electron (e^-)** → Discovered by J.J. Thomson in 1897 using cathode ray tube.
- **Proton (p^+)** → Discovered by Ernest Rutherford in 1919.
- **Neutron (n^0)** → Discovered by James Chadwick in 1932.

2. STRUCTURE OF ATOM

- Atom has a central nucleus.
- Nucleus contains protons (positive) and neutrons (neutral).
- Electrons revolve around the nucleus in specific energy levels or shells.
- Most of the mass of an atom is concentrated in the nucleus.
- Atom is neutral as number of protons = number of electrons.

Key Points ★

$p^+ \rightarrow +1$ charge

$e^- \rightarrow -1$ charge

$n^0 \rightarrow$ No charge

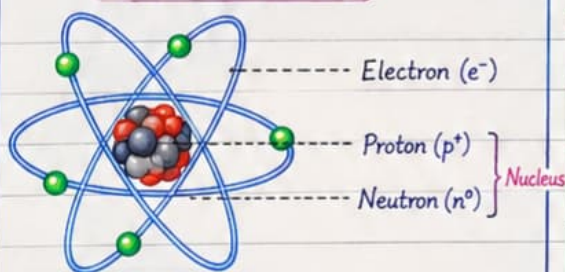
Mass →

$p^+ \approx 1$ amu

$n^0 \approx 1$ amu

$e^- \approx 1/1836$ amu

3D MODEL OF ATOM



Composition of Atom

Particle	Symbol	Charge	Mass (amu) (approx.)	Location
Electron	e^-	-1	$1/1836$	Outside nucleus
Proton	p^+	+1	1	Inside nucleus
Neutron	n^0	0	1	Inside nucleus

3. ATOMIC NUMBER (Z) & MASS NUMBER (A)

- **Atomic Number (Z)** → Number of protons in the nucleus.
– It is equal to number of electrons in a neutral atom.
- **Mass Number (A)** → Total number of protons + neutrons in the nucleus.

$$A = Z + N$$

For example :

For ${}^{14}_7\text{N}$ (Nitrogen)

$Z = 7$ (protons)

$A = 14$ (mass number)

$N = A - Z = 14 - 7 = 7$ (neutrons)

4. ISOTOPES

- Atoms of the same element having same atomic number (Z) but different mass numbers (A) are called isotopes.
- They have same chemical properties but different physical properties.

Example :

Hydrogen isotopes

${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$

(Protium) (Deuterium) (Tritium)



Remember : Structure of atom helps us understand periodic properties, chemical bonding and behaviour of elements.





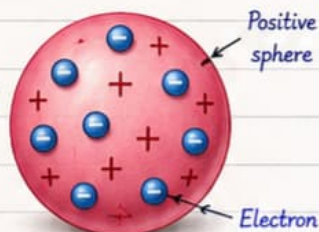
Structure of Atom



5. ATOMIC MODELS

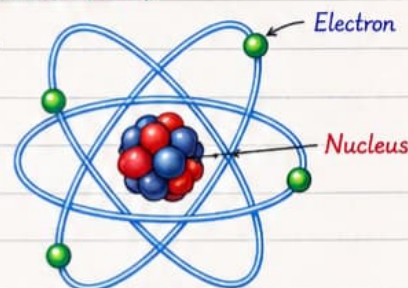
(a) Thomson's Model (1904)

- Atom is a sphere of positive charge.
- Electrons are embedded in it like raisins in a pudding.

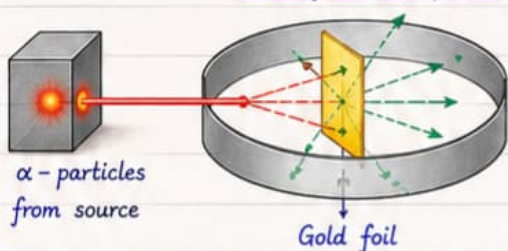


(b) Rutherford's Model (1911)

- Atom has a small, positively charged nucleus.
- Most of the space is empty.
- Electrons revolve around the nucleus.



Rutherford's α -particle Scattering Experiment



Most α -particles $\rightarrow$ passed straight (no deflection)

Some α -particles $\rightarrow$ deflected (small angle)

Very few α -particles $\rightarrow$ reflected back

Conclusion : Atom has dense positive nucleus with most of the space empty.

6. BOHR'S MODEL OF ATOM (1913)

- Electrons revolve around nucleus in fixed circular orbits called shells.
- Only those orbits are allowed for which electron's angular momentum is quantized.

$$mvr = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$$

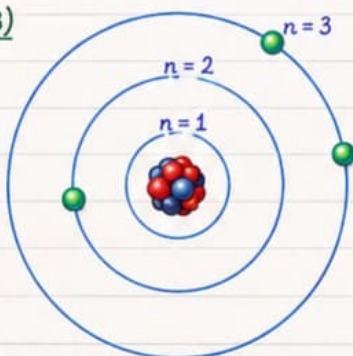
Where,

m = mass of electron

v = velocity

r = radius of orbit

h = Planck's constant
(6.626×10^{-34} Js)



Energy levels (n)

- $n = 1 \rightarrow$ K shell (closest)
 $n = 2 \rightarrow$ L shell
 $n = 3 \rightarrow$ M shell
 $n = 4 \rightarrow$ N shell (farthest)

Important Points

- ★ Only certain discrete orbits are allowed.
- ★ No energy is radiated in a stable orbit.
- ★ Energy is absorbed or emitted only during transition of electron from one orbit to another.

Energy of photon

$$h\nu = |E_2 - E_1|$$

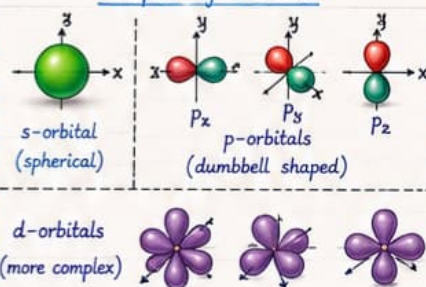
ν = frequency of radiation

7. QUANTUM NUMBERS

The exact position and energy of electron in an atom are described by four quantum numbers.

Quantum Number	Symbol	Possible Values	It describes	Example
1. Principal Quantum Number	n	1, 2, 3, 4	Main energy level or shell	$n = 1$ (K), 2 (L), 3 (M)
2. Azimuthal Quantum Number	l	0 to $(n-1)$	Subshell or shape of orbital	For $n=3 \rightarrow l = 0, 1, 2$ (s, p, d)
3. Magnetic Quantum Number	m_l	$-l$ to 0 to $+l$	Orientation of orbital in space	For $l=1 \rightarrow m_l = -1, 0, +1$
4. Spin Quantum Number	m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Direction of spin of electron	$m_s = +\frac{1}{2}$ ($\uparrow$), $-\frac{1}{2}$ ($\downarrow$)

Shapes of orbitals

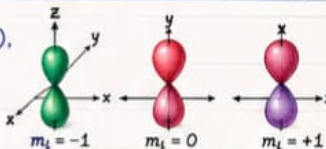


Note : Bohr's model works well for hydrogen like species (one electron atoms/ions) only.



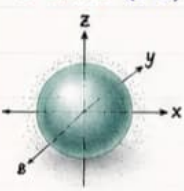
Structure of Atom

8. QUANTUM NUMBERS - describe the state of an electron

Quantum Number	Symbol	Possible Values	What it describes	Diagram / Example
1. Principal quantum number	n	1, 2, 3, 4 ...	Main energy level or shell	 $n=1 \rightarrow K$ $n=2 \rightarrow L$ $n=3 \rightarrow M$ $n=4 \rightarrow N \dots$
2. Azimuthal quantum number	l	0 to $(n-1)$	Subshell or shape of orbital	$l=0 \rightarrow s$ (spherical) $l=1 \rightarrow p$ (dumbbell) $l=2 \rightarrow d$ (clover leaf) $l=3 \rightarrow f$ (complex)
3. Magnetic quantum number	m_l	$-l$ to 0 to $+l$ ($2l+1$ values)	Orientation of orbital in space	For $l=1$ (p-orbital), $m_l = -1, 0, +1$ 
4. Spin quantum number	m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Direction of spin of electron	 $+\frac{1}{2}$ (spin up) $-\frac{1}{2}$ (spin down)

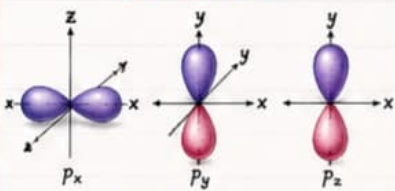
9. SHAPES OF ATOMIC ORBITALS

s-orbital ($l=0$)



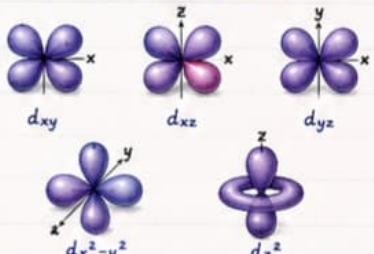
Spherical shape

p-orbitals ($l=1$)



Dumbbell shape

d-orbitals ($l=2$)



f-orbitals ($l=3$)



More complex (shapes)

★ Number of orbitals in a subshell = $2l+1$

So, $s = 1$ orbital, $p = 3$ orbitals, $d = 5$ orbitals, $f = 7$ orbitals

10. ELECTRONIC CONFIGURATION

- Distribution of electrons in various shells and subshells of an atom is called electronic configuration.
- Aufbau principle** : Electrons occupy orbitals in order of increasing energy (lowest to highest).

(a) Aufbau order (increasing energy)



Order :

$1s < 2s < 2p < 3s < 3p < 4s <$
 $3d < 4p < 5s < 4d < 5p <$
 $6s < 4f < 5d < 6p < 7s <$
 $5f < 6d < 7p \dots$

★ Orbitals with lower $(n+l)$ value have lower energy.

★ If $(n+l)$ is same, orbital with lower n has lower energy.

(b) Maximum electrons in a subshell

Subshell	s	p	d	f
No. of orbitals	1	3	5	7
Max. electrons	2	6	10	14

Each orbital can accommodate maximum 2 electrons with opposite spin.

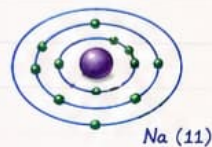
$\uparrow\downarrow$ = One orbital (max. 2 electrons)

(c) Examples

- For Sodium ($Z = 11$)

Electron configuration : $1s^2 2s^2 2p^6 3s^1$

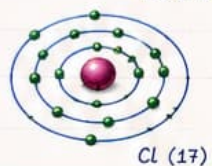
Shell wise distribution : $K = 2, L = 8, M = 1$



- For Chlorine ($Z = 17$)

Electron configuration : $1s^2 2s^2 2p^6 3s^2 3p^5$

Shell wise distribution : $K = 2, L = 8, M = 7$



Quick Summary

- ✓ Atom has nucleus (protons + neutrons) and electrons around it.
- ✓ Electrons occupy definite energy levels (shells) and orbitals (subshells).
- ✓ Quantum numbers describe the state of an electron completely.
- ✓ Electronic configuration follows Aufbau principle and Hund's rule.

"Understand the structure,
Master the nature !"





Structure of Atom

Class : 11th (NEET UG)

Subject : Chemistry

Topic : Structure of Atom

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11. ISOTOPES, ISOBARS & ISOTONES

★ Isotopes :

Atoms of same element having same atomic number (Z) but different mass number (A).

Example :

^1_1H , ^2_1H , ^3_1H → Isotopes of Hydrogen



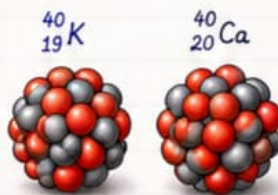
● = Proton (p^+) ● = Neutron (n^0)

★ Isobars :

Atoms of different elements having same mass number (A) but different atomic number (Z).

Example :

$^{40}_{19}\text{K}$ and $^{40}_{20}\text{Ca}$ → Isobars ($A = 40$)

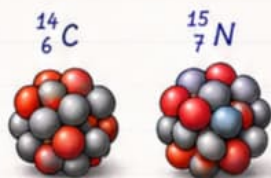


★ Isotones :

Atoms of different elements having same number of neutrons (N) but different atomic number (Z).

Example :

$^{14}_6\text{C}$ and $^{15}_7\text{N}$ → Isotones ($N = 8$)



RELATIONS BETWEEN Z, A & N

Mass number (A) = Atomic number (Z) + Number of neutrons (N)

$$A = Z + N$$

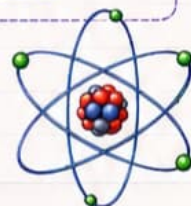
$Z \rightarrow$ Protons = Number of electrons (in a neutral atom)

$A \rightarrow$ Protons + Neutrons

$N \rightarrow$ Neutrons = $A - Z$

Significance

- ✓ Helps in identification of an element.
- ✓ Useful in writing electronic configuration.
- ✓ Explains stability, radioactivity and natural abundance of elements.

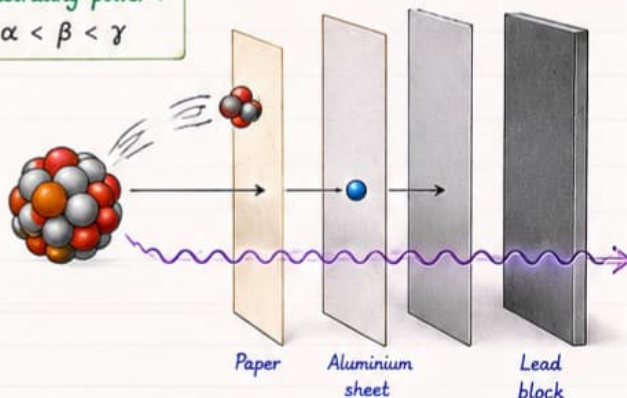


12. RADIOACTIVITY (Brief)

- Some nuclei are unstable and disintegrate spontaneously to give radiation.
- Types of radiation :

Radiation	Symbol	Nature	Charge	Penetrating Power	Stopped by
Alpha	α	Helium nucleus	+2	Low	Paper / Skin
Beta	β	Fast electron	-1	Medium	Aluminium sheet
Gamma	γ	High energy electromagnetic radiation	0	Very high	Lead / Concrete

Penetrating power :
 $\alpha < \beta < \gamma$



13. LIMITATIONS OF BOHR'S MODEL

- Failed to explain spectra of multi-electron atoms.
- Could not explain fine structure of hydrogen spectrum.
- Did not consider dual nature (wave-particle) of electron.
- Failed to account for intensity of spectral lines.

Modern view :

Quantum mechanical model (Schrödinger wave equation) overcomes these limitations.



14. IMPORTANT FACTS

- ★ The smallest possible value of n is 1.
- ★ Number of shells occupied in an atom = Period number in periodic table.
- ★ Valence electrons determine chemical behaviour.
- ★ Noble gases have stable configuration (complete octet/duplet).



QUICK FORMULAS

- ★ $A = Z + N$
- ★ $N = A - Z$
- ★ In neutral atom : $Z = e^-$
- ★ Max electrons in shell = $2n^2$
- ★ Orbitals in subshell = $2\ell + 1$
- ★ Max electrons in subshell = $2(2\ell + 1)$

QUICK SUMMARY

- ✓ Atom = Nucleus + Electrons
- ✓ Electrons occupy shells & subshells
- ✓ Quantized energy levels (Bohr's model)
- ✓ Quantum numbers describe electron
- ✓ Isotopes, Isobars, Isotones → key relations
- ✓ Radioactivity → α , β , γ radiations



NEET TIP

Learn concepts with diagrams, tables and formulas.
Practice NCERT examples and previous year questions regularly.



Remember : Structure of atom is the foundation of all chemistry.
Master it well for strong conceptual base in NEET !



All the Best for NEET !

